Detecting Artistic Influence Using Computer Vision: A Computational Analysis of African Mask Artworks Influence in Pablo Picasso's Paintings

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Abstract

Art historians have long observed that lot of paintings of Pablo Picasso show notable visual similarities to traditional African masks. While this influence has often been discussed through qualitative visual analysis and historical interpretation, the extent of this resemblance has rarely been examined using quantitative methods. This study proposes a computer vision–based approach to measure the visual similarity between African mask artworks and the facial forms represented in Picasso's paintings. To conduct this research, three famous Picasso Black Period painting images were taken and two African mask images were collected from publicly available museum collections. Visual features were extracted using two pre-trained convolutional neural networks (CNN), ResNet and CLIP, which generated feature embeddings that capture structural and stylistic patterns within each image. These embeddings were then compared using cosine similarity to quantify the degree of resemblance between the mask images and the painted faces. The results of the analysis produced a cosine similarity score of around 50 percent, indicating a measurable visual relationship between certain African mask structures and the geometric facial forms depicted in the paintings. These findings provide computational support for the long-standing art historical claim that African sculptural aesthetics influenced Picasso's artistic development and served as an important inspiration in the creation of these groundbreaking works.

Keywords: Computer vision, Convolutional Neural Networks, Cosine Similarity, African Art, Picasso, ResNet, CLIP

1 Introduction

The early 20th century marked a pivotal moment in the evolution of modern art, as European artists were looking for new forms of expression that challenged conventional aesthetics and representation [1]. Among these innovators, Pablo Picasso stands out for his radical transformations of form and perspective, particularly during his so-called "Black Period"

and the development of *Les Demoiselles d'Avignon* (1907), often regarded as the genesis of Cubism [1]. Art historians have long argued that Picasso's exposure to African sculptures and masks–encountered in Parisian ethnographic collections–profoundly influenced his stylistic shift toward abstraction, simplification, and fragmentation of the human figure [2, 3]. Yet, while these claims have been widely debated and visually demonstrated in art historical literature, the relationship between African sculptural forms and Picasso's paintings has rarely been examined through systematic, quantitative analysis [2, 3].

Recent advances in computer vision and machine learning, particularly in the use of convolutional neural networks (CNNs), have opened new possibilities for analyzing visual art in more objective and data-driven ways [8]. By transforming images into numerical representations, these models can detect patterns, shapes, and stylistic features at multiple levels of abstraction, offering a means to measure visual similarity across different artistic traditions [8]. Such techniques have already proven useful in tasks such as artwork classification, style recognition, and influence detection among painters, suggesting that computational methods can complement traditional art historical interpretation rather than replace it [4, 8]. Applying these approaches to the comparison between African mask art and Picasso's paintings allows for a new computational way of seeing similarities and influence between art forms.[2, 3].

This study aims to explore the similarities between African masks and Picasso's Black Period portraits through a computer vision–based approach. Using pretrained neural networks–Residual Neural Network (ResNet) and Contrastive Language-Image Pre-training (CLIP)–this research extracts high-dimensional visual features from both Picasso's paintings and selected African masks, and compares them using cosine similarity metrics to estimate degrees of visual resemblance. By quantifying structural and stylistic affinities between these images, the project seeks to provide computational evidence that can support, nuance, or challenge long-standing assertions about the role of African sculptural aesthetics in shaping Picasso's modernist innovations, particularly in the development of the geometric, mask-like facial forms that characterize this phase of his work [1, 2, 8].

The rest of the paper is organized as follows. Section 2 presents a review of related work on computational approaches to art analysis and on African influence in Picasso's work. Section 3 describes the proposed methodology, which is divided into three subsections: Image Collection, Feature Extraction Using ResNet and CLIP, and Similarity Measurement. The cosine similarity results obtained from both the ResNet and CLIP models are presented and analyzed in Section 4, covering the similarity scores for each painting–mask pair. Section 5 discusses the findings in the context of art-historical claims and addresses the limitations of the computational approach. Future directions for expanding this line of research are outlined in Section 6. Finally, concluding remarks are provided in Section 7.

2 Related Work

Scholars have extensively examined the influence of African art on *Les Demoiselles d'Avignon* and on Picasso's broader development, often emphasizing the role of masks and sculptural

forms in his move toward angular, fragmented faces and bodies. Joshua I. Cohen has published a comprehensive study detailing the influence of African mask artworks in Modern Art [7]. He describes how Picasso began collecting and studying African art for his paintings, and that influence can clearly be seen in Picasso's works during 1907–1909, also referred to as the Black Period [7, 6]. It is documented on Picasso's website that Picasso visited the Trocadéro museum for an exhibition and encountered African Art, yet he never gave it the credit it deserves for the influence on his works [7]. These accounts spark a debate on whether Picasso was actually influenced by African Art, which leads to the question of measuring that influence through a statistical approach — examining how similar the art forms Picasso created during that period are to African art forms.

In parallel, a growing body of work have used quantitative and computational methods to study art. Quantitative analyses of Picasso's creative process in works such as *Guernica* demonstrate how data and formal decomposition can illuminate artistic decisions over time [8]. Recent research on CLIP and related models has explored their ability to recognize artistic styles, relate images to text prompts, and measure stylistic adherence through cosine similarity in embedding space [5]. Other projects on artwork similarity search and vector representations show that CNN-based embeddings can capture both content and stylistic features, enabling image retrieval and influence analysis [4, 5]. However, to date, there appears to be little or no published work applying these techniques specifically to the question of African mask influence on Picasso's Black Period portraits, leaving a gap that this study begins to address.

3 Methodology

3.1 Image Collection

Three famous painting images created by Picasso during the Black Period were used in this study:

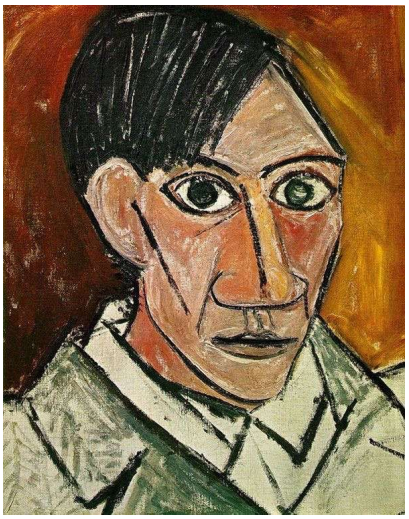
- **Painting 1:** *Les Demoiselles d'Avignon* (1907)
- **Painting 2:** *Self-Portrait*, oil on canvas, (1907)
- **Painting 3:** *Head of a Sleeping Woman (Study for Nude with Drapery)*, Paris, summer (1907)

Two African mask images were collected from publicly available museum archives:

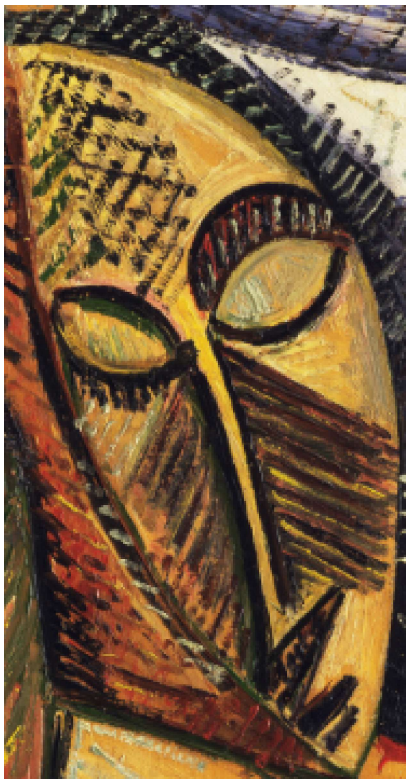
- **Face Mask 1:** Cleveland Museum of Art African face mask — `https://www.clevelandart.org/art/1953.456`
- **Face Mask 2:** Smithsonian African face mask — `https://www.si.edu/object/face-mask%3Anmafa_85-15-20`

The paintings and face masks used for this experiment can be seen in Fig 1 and Fig 2.

Painting 1: *Les Demoiselles d'Avignon* (1907)



Painting 2: *Self-Portrait* (1907)



Painting 3: *Head of a Sleeping Woman* (1907)

Figure 1: Picasso's Black Period Paintings used in this study.

Face Mask 1: African Face Mask from Cleveland Museum of Art

Face Mask 2: African Face Mask from Smithsonian

Figure 2: African Face Masks used in this study.

3.2 Feature Extraction Using ResNet and CLIP

To extract meaningful visual features, pre-trained convolutional neural networks (CNNs) were employed. ResNet50 was selected as a widely used architecture for visual feature representation, as it captures high-level patterns such as shapes, edges, and textures. CLIP (Contrastive Language–Image Pretraining) was additionally used due to its strong cross-modal representations and demonstrated sensitivity to artistic style [5].

ResNet50, introduced by He et al. [9], uses skip connections to capture hierarchical visual features such as edges, contours, and geometric shapes which makes it suitable for detecting structural similarities between the angular forms of African masks and the fragmented facial geometries in Picasso's Black Period paintings. CLIP, proposed by Radford et al. [10], is trained on 400 million image-text pairs and encodes broader stylistic and semantic properties beyond pure visual structure which was used to capture the overall aesthetic of the paintings used in the experiment. The model architecture for ResNet and CLIP can be seen in Fig 3 and Fig 4.

Therefore,instead of training the CNNs from scratch, transfer learning was applied. The final classification layer was removed from each network, and the models were used as feature extractors to generate high-dimensional embedding vectors representing each image. These embeddings encode structural and stylistic information learned from large-scale image datasets.

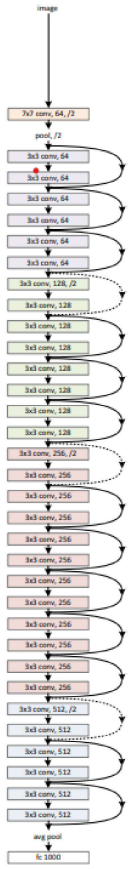


Figure 3: ResNet50 Model Feature Extraction Pipeline from [9]

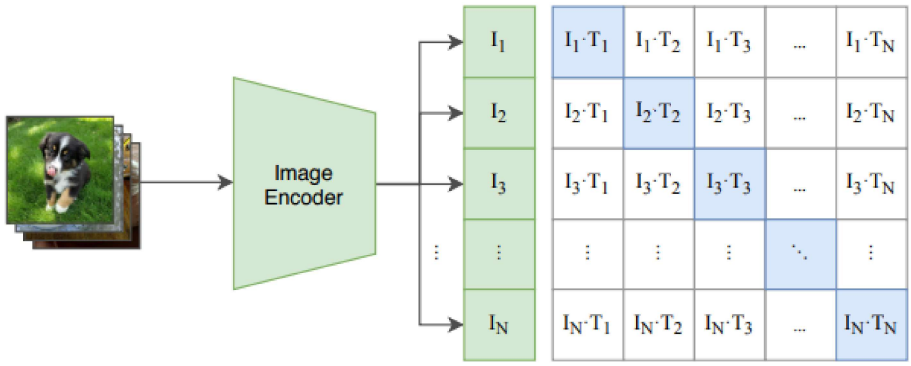


Figure 4: CLIP Feature Extraction Pipeline [10]

3.3 Similarity Measurement

For each African mask embedding, cosine similarity was computed against each painting embedding, separately for the ResNet and CLIP features. Cosine similarity measures the angle between two feature vectors in high-dimensional space. The score ranges from -1 to 1, where 1 indicates highly similar images, 0 indicates no similarity, and -1 indicates images with opposite patterns. Given two vectors $\mathbf{x}$ and $\mathbf{y}$, cosine similarity is defined as:

$$\text{cosine_similarity}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \, \|\mathbf{y}\|} \tag{1}$$

4 Results

4.1 ResNet Similarity Scores

Cosine similarity scores obtained from ResNet embeddings are summarized in Table 1.

Table 1: Cosine Similarity Between Picasso Paintings and African Masks (ResNet)

Painting	**Mask**	**Similarity Score**
Painting 1	Face Mask 1	0.2316
	Face Mask 2	0.2723
Painting 2	Face Mask 1	0.5454
	Face Mask 2	0.6076
Painting 3	Face Mask 1	0.3942
	Face Mask 2	0.4670

Overall, *Self-Portrait* (Painting 2) shows the highest similarity to both masks, with scores above 0.54, while *Les Demoiselles d'Avignon* (Painting 1) exhibits lower particularly to Face Mask 2.

4.2 CLIP Similarity Scores

Cosine similarity scores in the CLIP embedding space are reported in Table 2.

In CLIP, the similarity scores cluster in the 0.43–0.54 range and the Painting 3 (*Head of a Sleeping Woman*) showed the strongest resemblance to both masks. The scores suggest a consistent, moderate degree of alignment between the facial structures in these paintings and the African mask forms as encoded by CLIP.

5 Discussion

The results from both ResNet and CLIP show a clear visual connection between the selected African masks and Picasso's Black Period portraits. Several mask-painting pairs have cosine

Table 2: Cosine Similarity Between Picasso Paintings and African Masks (CLIP)

Painting	**Mask**	**Similarity Score**
Painting 1	Face Mask 1	0.4615
	Face Mask 2	0.4477
Painting 2	Face Mask 1	0.4351
	Face Mask 2	0.4328
Painting 3	Face Mask 1	0.5038
	Face Mask 2	0.5386

similarity values that suggest a moderate alignment in feature space. In ResNet, the strongest ties appear between Self-Portrait and both masks. In CLIP, Head of a Sleeping Woman is most closely aligned with the mask embeddings. These differences likely show the unique training goals and biases of the two models.

From an art-historical perspective, these computational findings support long-standing observations that Picasso's Black Period works feature mask-like facial shapes. This includes long features, sharp outlines, and flat shapes that remind us of African masks. The moderate similarity scores do not mean that the paintings copy specific masks directly. Instead, they indicate that the formal structures of the paintings share a related area of visual feature space. This supports the idea of stylistic inspiration and transformation rather than direct imitation. It fits with research that views Picasso's connection to African art as both transformative and intertwined with colonial appropriation[**?**].

At the same time, the study is limited as well because only two face masks and three paintings were taken instead of the whole dataset and the models were not also trained solely on Picasso's paintings images. As only face masks were taken, it excluded other culturally significant features such as bodies, nudity, and spatial composition influenced by other African artworks at the time. Cosine similarity in a general image-embedding space also doesn't consider historical context, symbolic meaning, or the ethical aspects of appropriation. It only measures visual similarities based on specific models. Therefore, these results should be seen as an initial quantitative addition to critical art-historical interpretation, not a substitute for it.

6 Future Directions

Future work can grow this work by collecting more of picassos art works and African masks from different sources making a big dataset. A special CNN model could learn from these art collections to pick up small style differences in early modernism and African art. Also, mixing computer results with old records, postcolonial ideas, and museum views might show how visual ideas, culture shifts, and power play out in modern art history could give a fuller picture [4].

7 Conclusion

This study introduced a computer vision–based framework to investigate the visual relationship between African mask aesthetics and Picasso's Black Period paintings. Using pretrained ResNet and CLIP models as feature extractors and cosine similarity as a similarity metric, the analysis found consistent, moderate levels of similarity between the facial structures in three Picasso paintings and two African face masks from museum collections. These quantitative results lend computational support to art-historical claims that African sculptural forms influenced Picasso's development of geometric, mask-like facial treatments during this period.

References

[1] X. Kang, "On the Way to Three-dimension—'Les Demoiselles D'Avignon' by Picasso," Art and Society, vol. 2, no. 4, 2021. doi: 10.32629/asc.v2i4.523.

[2] M. Hummer, "The Appropriation of African Objects in Pablo Picasso's Les Demoiselles d'Avignon," 2018. [Online]. Available: `https://www.academia.edu/99238585/The_Appropriation_of_African_Objects_in_Pablo_Picasso_s_Les_Demoiselles_d_Avignon`. [Accessed: Mar. 8, 2026].

[3] Dr. Peri Klemm, "The reception of African art in the West," in Smarthistory, December 20, 2016, accessed March 9, 2026, https://smarthistory.org/the-reception-of-african-art-in-the-west/.

[4] O. Boughaba, "Artwork Similarity Search – Exploring the Power of Vector Databases," Jun. 29, 2023. [Online]. Available: https://otmaneboughaba.com/posts/artwork-similarity-search/. [Accessed: Mar. 9, 2026].

[5] Asperti, A., Dessi, L., Tonetti, M. C., Wu, N. (2025, October). Does CLIP perceive art the same way we do?. In 2025 International Conference on Content-Based Multimedia Indexing (CBMI) (pp. 1-8). IEEE.

[6] "Picasso's African Period," PabloPicasso.org. [Online]. Available: `https://www.pablopicasso.org/africanperiod.jsp`

[7] J. I. Cohen, *Masks and the Modern: African/European Encounters in 20th-Century Art.* Columbia University, 2012.

[8] R. W. Weisberg, "On Structure in the Creative Process: A Quantitative Case-Study of the Creation of Picasso's Guernica," *Empirical Studies of the Arts*, vol. 22, no. 1, pp. 23–54, 2004. doi: 10.2190/EH48-K59C-DFRB-LXE7.

[9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[10] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, and I. Sutskever, "Learning Transferable Visual Models from Natural Language Supervision," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 8748–8763.